Hydrocarbons

Compounds of carbon and hydrogen alone, and the fuels, polymers, solvents, dyes and drugs that start from them.

By the end of this unit

- give a hydrocarbon its IUPAC name
- pick out and draw the isomers of alkanes, alkenes, alkynes and aromatic hydrocarbons
- know the routes by which hydrocarbons are prepared
- tell alkanes, alkenes, alkynes and aromatics apart by how they look and how they react
- draw the conformations of ethane and say how they differ
- see what hydrocarbons do for us as energy sources and in industry
- work out, from the electronic mechanism, what an unsymmetrical alkene or alkyne adds to give
- grasp the structure of benzene, what aromaticity means, and how benzene undergoes electrophilic substitution
- foresee where a substituent already on a benzene ring will send the next one
- know which hydrocarbons are carcinogenic or toxic

Hydrocarbons in the day's routine

- the word says it: a **hydrocarbon** holds **carbon and hydrogen, nothing else**
- **LPG** is **liquified petroleum gas**, **CNG** is **compressed natural gas**, and **LNG**, much in the news, is **liquified natural gas**, a fuel made by liquifying natural gas
- **petrol, diesel and kerosene oil** come off the **fractional distillation of petroleum** from under the earth's crust; **coal gas** comes from the **destructive distillation of coal**
- **natural gas** turns up in the **upper strata while oil wells are drilled**; compressed, it is CNG
- **LPG** is the **domestic fuel that pollutes least**; **kerosene oil** serves at home too, but **pollutes somewhat**
- cars run on **petrol, diesel and CNG**; the **petrol and CNG ones pollute less**
- each of these fuels is a **mixture of hydrocarbons**, and each is a **source of energy**

Beyond fuel

- hydrocarbons go into **polymers** such as **polythene, polypropene and polystyrene**
- the **higher hydrocarbons are paint solvents**, and the **starting point for many dyes and drugs**
- so they matter in daily life on every side, and this unit takes them up in detail

A.1 Classification

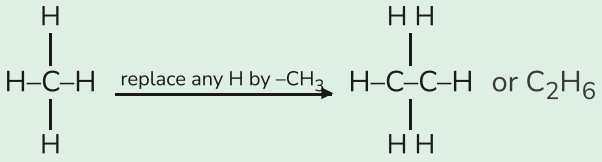
- hydrocarbons come in several kinds; going by the **carbon-carbon bonds they hold**, there are **three main categories**, the first being the **(i) saturated** hydrocarbons — continued on sheet 2

The unit runs in four parts: **A · Classification and alkanes**, then **B · Alkenes**, then **C · Alkynes**, then **D · Aromatic hydrocarbons and toxicity**.

- the three categories, then: **(i) saturated, (ii) unsaturated, (iii) aromatic** hydrocarbons
- a **saturated** hydrocarbon has **C–C and C–H single bonds only**; carbons strung into an **open chain** by single bonds make an **alkane**, met in Unit 8, and carbons **closed into a ring** make a **cycloalkane**
- an **unsaturated** hydrocarbon carries **C–C multiple bonds**, double, triple, or both
- **aromatic** hydrocarbons are a **special kind of cyclic compound**
- with carbon **tetravalent** and hydrogen **monovalent**, one can build any number of models of both open and closed chains: for alkanes, **plasticine balls as atoms joined by toothpicks as bonds** will do, and **spring models** for alkenes, alkynes and aromatics

A.2 Alkanes

- alkanes are the **saturated open chain hydrocarbons**, all their **C–C bonds single**
- the first of the family is **methane (CH_4)**, a gas that **coal mines and marshes give off**
- swap one hydrogen of methane for a carbon, give that carbon the hydrogens its tetravalence asks for, and the result is **C_2H_6, ethane**: methane with an **H replaced by $–CH_3$**
- keep replacing H by $–CH_3$ and the series runs on: **C_3H_8, C_4H_{10} …**



- under ordinary conditions alkanes are **inert**, unmoved by **acids, bases and other reagents**; hence the old name **paraffins** (Latin *parum*, little; *affinis*, affinity)
- set the formulas side by side and the **general formula** of the family, its homologous series, comes out as **C_nH_{2n+2}**; put in the right *n* and it gives any homologue
- by **VSEPR theory** (Unit 4) methane is a **tetrahedron** (Figure 9.A): **carbon at the centre, four hydrogens at the corners**, every **H–C–H angle 109.5°**
- an alkane is **tetrahedra joined together**, with **C–C 154 pm** and **C–H 112 pm** (Unit 8)
- the **C–C and C–H σ bonds** come of **head-on overlap** of carbon ***sp*^{3}** orbitals with each other and with hydrogen **1*s***

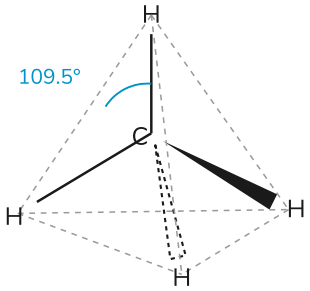


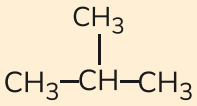
Figure 9.A Structure of methane

Nomenclature and isomerism

- Unit 8 covered how the classes of organic compound are named; a few more alkane examples take naming and isomerism further, with **common names in parenthesis**
- **methane, ethane and propane** each have **one structure only**; from butane up there is **more than one**
- the four carbons of **C_4H_{10}** join either in a **continuous chain** or a **branched** one, two ways in all

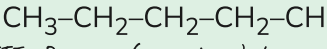
$$CH_3–CH_2–CH_2–CH_3$$

I · Butane (n-butane), b.p. 273 K

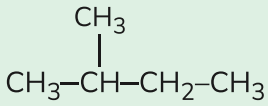


II · 2-Methylpropane (isobutane), b.p. 261 K

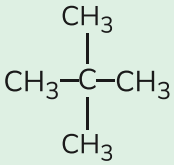
- **I and II share a molecular formula** yet **boil at different temperatures** and differ in other properties too
- five carbons and twelve hydrogens, **C_5H_{12}**, can be put together **three ways**, structures **III–V**, and again the three share the formula but **not the properties**
- I and II are the **isomers of butane**; III, IV and V the **isomers of pentane**
- the properties differ because the **structures** do, so these are **structural isomers**
- I and III have a **continuous carbon chain**, II, IV and V a **branched** one; structural isomers that differ in the **carbon chain** are **chain isomers**
- so **C_4H_{10} has two chain isomers and C_5H_{12} three**



III · Pentane (n-pentane), b.p. 309 K



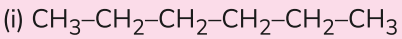
IV · 2-Methylbutane (isopentane), b.p. 301 K



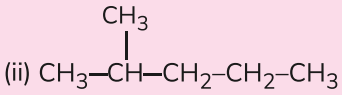
V · 2,2-Dimethylpropane (neopentane), b.p. 282.5 K

Worked example 9.a

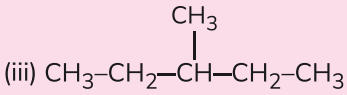
Draw every chain isomer of the alkane C_6H_{14}, and give each its IUPAC name.

(i) 

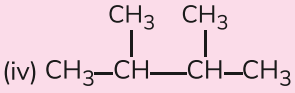
n-Hexane

(ii) 

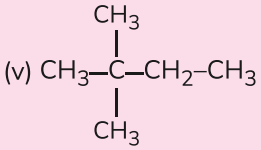
2-Methylpentane

(iii) 

3-Methylpentane

(iv) 

2,3-Dimethylbutane

(v) 

2,2-Dimethylbutane

Primary, secondary, tertiary, quaternary carbon

- a carbon is graded **primary (1°), secondary (2°), tertiary (3°) or quaternary (4°)** by **how many carbons are bonded to it**
- **primary**: bonded to **no other carbon**, as in methane, or to **one**, as in ethane; a **terminal carbon is always primary**
- **secondary**: bonded to **two** carbons · **tertiary**: to **three** · **neo or quaternary**: to **four**
- try marking the 1°, 2°, 3° and 4° carbons in the structures above — continued on sheet 4

- go on up the alkanes and the **isomer count climbs fast**: **C_6H_{14} has five, C_7H_{16} nine**, and **$C_{10}H_{22}$ as many as 75**
- in II, IV and V a **$–CH_3$ sits on carbon 2**; groups like **$–CH_3$, $–C_2H_5$, $–C_3H_7$** turn up on carbon in alkanes and in other classes too

- such substituents are **alkyl groups**: an alkane **less one hydrogen**
- their general formula is **C_nH_{2n+1}** (Unit 8)
- the naming rules of Unit 8 still apply; the example below shows them at work on substituted alkanes

Worked example 9.b

Draw each isomeric alkyl group of formula C_5H_{11}. Then attach –OH to the open end in each case, and name the alcohol that results by the IUPAC system.

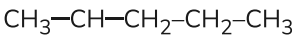
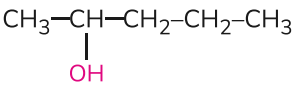
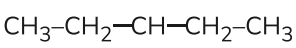
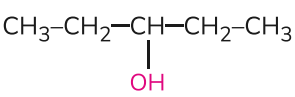
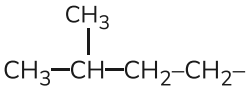
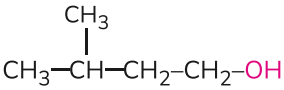
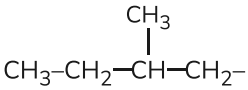
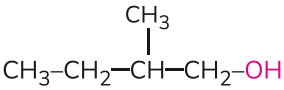
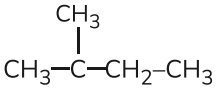
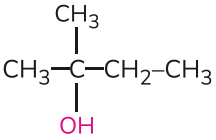
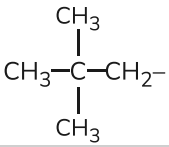
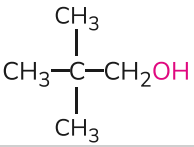
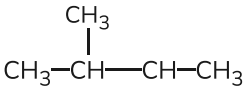
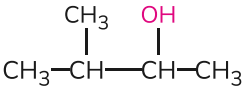
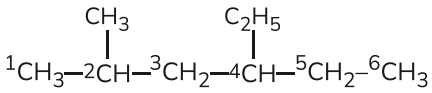
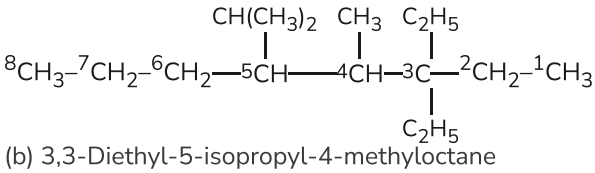
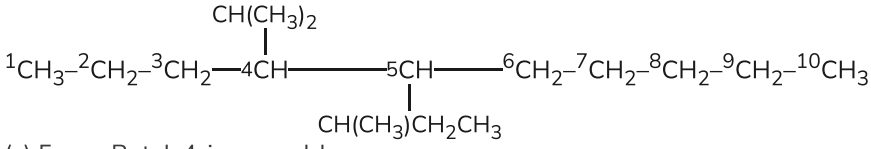
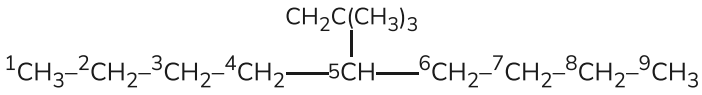
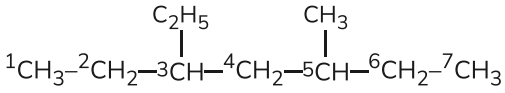
	–C_5H_{11} group	*Alcohol*	*Name*
(i)	$CH_3–CH_2–CH_2–CH_2–CH_2–$	$CH_3–CH_2–CH_2–CH_2–CH_2–OH$	Pentan-1-ol
(ii)			Pentan-2-ol
(iii)			Pentan-3-ol
(iv)			3-Methylbutan-1-ol
(v)			2-Methylbutan-1-ol
(vi)			2-Methylbutan-2-ol
(vii)			2,2-Dimethylpropan-1-ol
(viii)			3-Methylbutan-2-ol

Table 9.A Nomenclature of a Few Organic Compounds

Structure and IUPAC Name	Remarks
 (a) 4-Ethyl-2-methylhexane	the locant sum is kept lowest, and the groups run alphabetically
 (b) 3,3-Diethyl-5-isopropyl-4-methyloctane	lowest sum again, then alphabetical order
 (c) 5-sec-Butyl-4-isopropyldecane	sec is passed over when alphabetising; isopropyl counts as one word
 (d) 5-(2,2-Dimethylpropyl)nonane	the side chain's own substituents get a numbering of their own
 (e) 3-Ethyl-5-methylheptane	alphabetical order sets the priority

Worked example 9.c

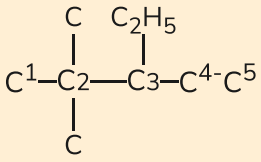
Name by the IUPAC system: (i) $(CH_3)_3CCH_2C(CH_3)_3$; (ii) $(CH_3)_2C(C_2H_5)_2$; (iii) tetra-*tert*-butylmethane.

- (i) **2,2,4,4-Tetramethylpentane**
- (ii) **3,3-Dimethylpentane**
- (iii) **3,3-Di-*tert*-butyl-2,2,4,4-tetramethylpentane**

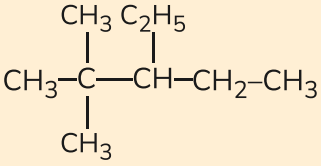
From name back to structure

- naming a structure correctly matters; **drawing the right structure from an IUPAC name matters just as much**
- first write out the **longest chain**, the one the parent alkane stands for; **number it**; hang the **substituents on the right carbons**; last, **fill each carbon's valence with hydrogens**
- for **3-ethyl-2,2-dimethylpentane** the steps run:
- **i)** draw the five-carbon chain: **C–C–C–C–C**
- **ii)** number it: **C^1–C^2–C^3–C^4–C^5** — continued on sheet 6

- **iii)** put the **ethyl on carbon 3** and **two methyls on carbon 2**



- **iv)** give every carbon the **hydrogens its valence needs**

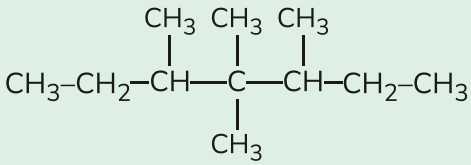


- and the structure is right; the two examples that follow test the same skill

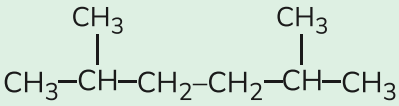
Worked example 9.d

Draw (i) 3,4,4,5-Tetramethylheptane and (ii) 2,5-Dimethylhexane.

(i)



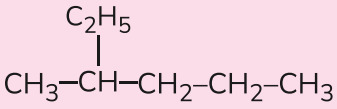
(ii)



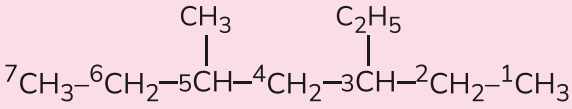
Worked example 9.e

Draw each compound. Then say what is wrong with the name given, and supply the correct IUPAC name: (i) 2-Ethylpentane; (ii) 5-Ethyl-3-methylheptane.

(i)



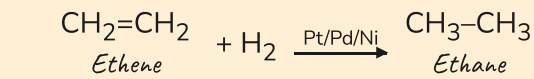
(ii)

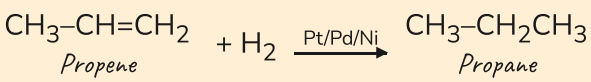


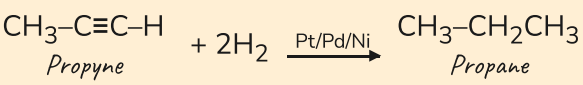
- the **longest chain runs to six carbons**, not five, so the name is **3-Methylhexane**
- **number from the end that gives ethyl the lower locant**; so **3-ethyl-5-methylheptane** it is

Preparation

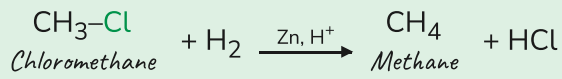
- most alkanes are drawn from **petroleum and natural gas**; the laboratory has these routes
- **1 · From unsaturated hydrocarbons — dihydrogen adds across an alkene or alkyne** over a **finely divided metal, platinum, palladium or nickel**, and an alkane results; the process is **hydrogenation**
- the metal **adsorbs H_2 on its surface and activates the H–H bond**; **Pt and Pd work at room temperature**, **Ni needs higher temperature and pressure**







- **2 · From alkyl halides — i)** an alkyl halide, **fluorides excepted**, is **reduced by zinc and dilute hydrochloric acid** to the alkane



$$\underset{\text{Chloroethane}}{C_2H_5{-}Cl} + H_2 \xrightarrow{Zn,\ H^+} \underset{\text{Ethane}}{C_2H_6} + HCl$$

$$\underset{\text{1-Chloropropane}}{CH_3CH_2CH_2Cl} + H_2 \xrightarrow{Zn,\ H^+} \underset{\text{Propane}}{CH_3CH_2CH_3} + HCl$$

- **ii)** an alkyl halide treated with **sodium metal in dry ether**, no moisture about, gives a **higher alkane**: the **Wurtz reaction**, the route to **higher alkanes with an even number of carbons**

$$\underset{\text{Bromomethane}}{CH_3Br} + 2Na + \underset{}{BrCH_3} \xrightarrow{\text{dry ether}} \underset{\text{Ethane}}{CH_3{-}CH_3} + 2NaBr$$

$$\underset{\text{Bromoethane}}{C_2H_5Br} + 2Na + \underset{}{BrC_2H_5} \xrightarrow{\text{dry ether}} \underset{\text{n-Butane}}{C_2H_5{-}C_2H_5} + 2NaBr$$

- what comes of taking **two different alkyl halides**? worth thinking through

- **3 · From carboxylic acids — i)** the **sodium salt of a carboxylic acid, heated with soda lime** (sodium hydroxide plus calcium oxide), gives an alkane with **one carbon fewer than the acid**
- knocking CO_2 out of a carboxylic acid this way is **decarboxylation**

$$\underset{\text{Sodium ethanoate}}{CH_3COO^-Na^+} + NaOH \xrightarrow[\Delta]{CaO} CH_4 + Na_2CO_3$$

Worked example 9.f

To make propane by this route, which acid's sodium salt is wanted? Give the equation.

- **butanoic acid**:

$$CH_3CH_2CH_2COO^-Na^+ + NaOH \xrightarrow{CaO} CH_3CH_2CH_3 + Na_2CO_3$$

- **ii) Kolbe's electrolysis — electrolyse an aqueous solution of the sodium or potassium salt** of a carboxylic acid, and an **alkane with an even number of carbons** comes off **at the anode**

$$\underset{\text{Sodium acetate}}{2CH_3COO^-Na^+ + 2H_2O} \xrightarrow{\text{Electrolysis}} CH_3{-}CH_3 + 2CO_2 + H_2 + 2NaOH$$

- the path is thought to run:

i) $2CH_3COO^-Na^+ \longrightarrow 2CH_3{-}COO^- + 2Na^+$

ii) at anode: $\underset{\text{Acetate ion}}{2CH_3{-}COO^-} \xrightarrow{-2e^-} \underset{\text{Acetate free radical}}{2CH_3{-}COO\bullet}$

$\longrightarrow \underset{\text{Methyl free radical}}{2\dot{C}H_3} + 2CO_2\uparrow$

iii) $H_3C\bullet + \bullet CH_3 \longrightarrow H_3C{-}CH_3\uparrow$

iv) at cathode: $H_2O + e^- \longrightarrow {}^-OH + H$; $H + H \longrightarrow H_2\uparrow$

- **methane cannot be made this way**: why not?

Properties · physical

- alkanes are **all but non-polar**: their **C–C and C–H bonds are covalent**, and **carbon and hydrogen differ very little in electronegativity**
- only **weak van der Waals forces** hold them together, so at 298 K **C_1 to C_4 are gases, C_5 to C_{17} liquids**, and **from 18 carbons up, solids**; all are **colourless and odourless**
- being non-polar, how well would they **dissolve in water**?
- **petrol**, a hydrocarbon mixture, **fuels cars**; petrol and the **lower petroleum fractions** also **dry-clean clothes, lifting grease stains**
- that tells us something about grease: it is a **mixture of higher alkanes**, and so **non-** — continued on sheet 8

- —polar, and so **hydrophobic**
- the general rule of solubility: **polar dissolves in polar, non-polar in non-polar**, that is **like dissolves like**
- Table 9.B lists boiling points, and they **climb steadily with molecular mass**: a **bigger molecule, or a larger surface, means stronger van der Waals forces** between molecules
- the **three isomeric pentanes** (pentane, 2-methylbutane, 2,2-dimethylpropane) repay a look: **pentane, with its unbroken five-carbon chain, boils highest, 309.1K**, while **2,2-dimethylpropane boils at 282.5K**
- **more branching pushes the molecule towards a sphere**; spheres **touch over less area**, the **forces between them are weaker**, and **less heat overcomes them**

Table 9.B Variation of Melting Point and Boiling Point in Alkanes

Molecular formula	Name	Molecular mass/u	b.p./(K)	m.p./(K)
CH_4	Methane	16	111.0	90.5
C_2H_6	Ethane	30	184.4	101.0
C_3H_8	Propane	44	230.9	85.3
C_4H_{10}	Butane	58	272.4	134.6
C_4H_{10}	2-Methylpropane	58	261.0	114.7
C_5H_{12}	Pentane	72	309.1	143.3
C_5H_{12}	2-Methylbutane	72	300.9	113.1
C_5H_{12}	2,2-Dimethylpropane	72	282.5	256.4
C_6H_{14}	Hexane	86	341.9	178.5
C_7H_{16}	Heptane	100	371.4	182.4
C_8H_{18}	Octane	114	398.7	216.2
C_9H_{20}	Nonane	128	423.8	222.0
$C_{10}H_{22}$	Decane	142	447.1	243.3
$C_{20}H_{42}$	Eicosane	282	615.0	309.7

Properties · chemical

- as said, alkanes **generally sit inert before acids, bases, oxidising and reducing agents**; under the right conditions, though, they do react
- **1 · Substitution reactions — one or more hydrogens** can be swapped for a **halogen, a nitro group or a sulphonic acid group**
- **halogenation** needs either **heat (573-773 K)** or **diffused sunlight or ultraviolet light**
- the **lower alkanes will not nitrate or sulphonate**
- reactions that **swap out an alkane's hydrogens** are **substitution reactions**; the **chlorination of methane** shows the pattern

Halogenation

$$CH_4 + Cl_2 \xrightarrow{h\nu} \underset{\textit{Chloromethane}}{CH_3Cl} + HCl$$

$$CH_3Cl + Cl_2 \xrightarrow{h\nu} \underset{\textit{Dichloromethane}}{CH_2Cl_2} + HCl$$

$$CH_2Cl_2 + Cl_2 \xrightarrow{h\nu} \underset{\textit{Trichloromethane}}{CHCl_3} + HCl$$

$$CHCl_3 + Cl_2 \xrightarrow{h\nu} \underset{\textit{Tetrachloromethane}}{CCl_4} + HCl$$

$$CH_3{-}CH_3 + Cl_2 \xrightarrow{h\nu} CH_3{-}CH_2Cl + HCl$$

Chloroethane

- the halogens react with alkanes in the order **F_2 > Cl_2 > Br_2 > I_2**, and the hydrogens are replaced in the order **3° > 2° > 1°**
- **fluorination runs too violently to control**; **iodination is very slow and reversible**, and is done with an **oxidising agent such as HIO_3 or HNO_3** present

$$CH_4 + I_2 \rightleftharpoons CH_3I + HI$$

$$HIO_3 + 5HI \longrightarrow 3I_2 + 3H_2O$$

- halogenation is believed to go by a **free radical chain mechanism** in **three steps: initiation, propagation, termination**

Mechanism

- **(i) Initiation** — **light or heat homolyses the chlorine molecule**; the **Cl–Cl bond is weaker than C–C or C–H**, so it is the one that gives first

$$Cl{-}Cl \xrightarrow[\text{homolysis}]{h\nu} \dot{Cl} + \dot{Cl}$$

Chlorine free radicals

- **(ii) Propagation** — the **chlorine radical attacks methane**, **breaks a C–H bond**, and drives the reaction on: a **methyl free radical** and **H–Cl** result

$$\text{(a)}\quad CH_4 + \dot{Cl} \xrightarrow{h\nu} \dot{C}H_3 + H{-}Cl$$

- that methyl radical **attacks a second chlorine molecule**, giving **CH_3–Cl** and, by homolysis of Cl_2, **a fresh chlorine radical**

$$\text{(b)}\quad \dot{C}H_3 + Cl{-}Cl \xrightarrow{h\nu} CH_3{-}Cl + \dot{Cl}$$

- the radicals so made **repeat (a) and (b) in turn, and a chain runs**
- (a) and (b) are the propagation steps that **lead straight to the main products**, but **other propagation steps can and do occur**; the two below show where the **more highly halogenated products** come from

$$CH_3Cl + \dot{Cl} \longrightarrow \dot{C}H_2Cl + HCl$$

$$\dot{C}H_2Cl + Cl{-}Cl \longrightarrow CH_2Cl_2 + \dot{Cl}$$

- **(iii) Termination** — in time the reaction **stops, the reactants used up** and/or **side reactions** setting in; the chain can end by

$$\text{(a)}\quad \dot{Cl} + \dot{Cl} \rightarrow Cl{-}Cl \qquad \text{(b)}\quad H_3\dot{C} + \dot{C}H_3 \rightarrow H_3C{-}CH_3$$

$$\text{(c)}\quad H_3\dot{C} + \dot{Cl} \rightarrow H_3C{-}Cl$$

- (c) does yield a product, CH_3–Cl, yet it **uses up radicals and so ends the chain**
- the mechanism also explains why **ethane turns up as a by-product** when methane is chlorinated

- **2 · Combustion** — **heated in air or dioxygen**, an alkane is **oxidised right through to carbon dioxide and water**, giving off a **great deal of heat**

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$
$$\Delta_cH^{\ominus} = -890\ \text{kJ mol}^{-1}$$

$$C_4H_{10}(g) + 13/2\ O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$$
$$\Delta_cH^{\ominus} = -2875.84\ \text{kJ mol}^{-1}$$

- for any alkane the combustion runs

$$C_nH_{2n+2} + \frac{3n+1}{2} O_2 \rightarrow nCO_2 + (n+1)H_2O$$

- that **outpouring of heat is why alkanes serve as fuels**
- burnt in **too little air or dioxygen**, an alkane gives **carbon black**, which goes into **ink, printer ink, black pigments**, and serves as **filters**

$$CH_4(g) + O_2(g) \xrightarrow[\text{combustion}]{\text{incomplete}} C(s) + 2H_2O(l)$$

- **3 · Controlled oxidation** — heat an alkane with a **measured supply of dioxygen or air, at high pressure, over the right catalyst**, and various **oxidation products** result

(i) $2CH_4 + O_2 \xrightarrow{Cu/523K/100atm} \underset{\textit{Methanol}}{2CH_3OH}$

(ii) $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} \underset{\textit{Methanal}}{HCHO} + H_2O$

(iii) $2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn} \underset{\textit{Ethanoic acid}}{2CH_3COOH} + 2H_2O$

- **(iv)** alkanes ordinarily **resist oxidation**, but one with a **tertiary hydrogen** is taken to the **alcohol by potassium permanganate**

$$\underset{\textit{2-Methylpropane}}{(CH_3)_3CH} \xrightarrow[\text{Oxidation}]{KMnO_4} \underset{\textit{2-Methylpropan-2-ol}}{(CH_3)_3COH}$$

- **6 · Reaction with steam** — **methane and steam at 1273 K over nickel** give **carbon monoxide and dihydrogen**; industry **makes its dihydrogen this way**

$$CH_4 + H_2O \xrightarrow[\Delta]{Ni} CO + 3H_2$$

- **7 · Pyrolysis** — the **higher alkanes break up on strong heating** into **lower alkanes, alkenes** and the like; **heat-driven decomposition into smaller fragments** is **pyrolysis**, or **cracking**

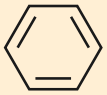
$$C_6H_{14} \xrightarrow{773K} \begin{matrix} C_6H_{12} + H_2 \\ C_4H_8 + C_2H_6 \\ C_3H_6 + C_2H_4 + CH_4 \end{matrix}$$

- pyrolysis of an alkane is thought to be a **free radical reaction**
- making **oil gas, or petrol gas, out of kerosene oil or petrol** rests on it

- **4 · Isomerisation** — heat an ***n*-alkane with anhydrous aluminium chloride and hydrogen chloride gas** and it **rearranges to branched alkanes**; the **major products** are shown, and **minor ones** are possible too, worth thinking out, though **organic reactions seldom report them**

$$\underset{\textit{n-Hexane}}{CH_3(CH_2)_4CH_3} \xrightarrow{\text{Anhy. } AlCl_3/HCl} \underset{\textit{2-Methylpentane}}{CH_3{-}\overset{\overset{CH_3}{|}}{CH}{-}(CH_2)_2{-}CH_3} + \underset{\textit{3-Methylpentane}}{CH_3CH_2{-}\overset{\overset{CH_3}{|}}{CH}{-}CH_2{-}CH_3}$$

- **5 · Aromatization** — an ***n*-alkane of six carbons or more**, heated to **773K at 10-20 atmospheres** over **oxides of vanadium, molybdenum or chromium on alumina**, is **dehydrogenated and closed into a ring**: **benzene and its homologues** result, by **aromatization**, also called **reforming**

$$\underset{\textit{n-Hexane}}{CH_3(CH_2)_4CH_3} \xrightarrow[773K,\ 10\text{–}20\ atm]{Cr_2O_3 \text{ or } V_2O_5 \text{ or } Mo_2O_3} \text{  } \longleftrightarrow \text{  } + 4H_2$$

- **toluene (C_7H_8)** is **methylbenzene**: which alkane would make it?
- **dodecane**, a constituent of kerosene oil, heated to **973K over Pt, Pd or Ni**, breaks into **heptane and pentene together**

$$\underset{\textit{Dodecane}}{C_{12}H_{26}} \xrightarrow[973K]{Pt/Pd/Ni} \underset{\textit{Heptane}}{C_7H_{16}} + \underset{\textit{Pentene}}{C_5H_{10}} + \text{Other products}$$

Conformations

- an alkane's **C–C bonds are σ bonds**, and a σ molecular orbital's **electron cloud is symmetrical about the internuclear axis**; **turning the bond about that axis disturbs nothing**
- so **rotation about a C–C single bond is free**, and it throws up **different arrangements of the atoms in space**, each turning into the next
- arrangements **interconvertible by rotation about a C–C single bond** are **conformations**, also **conformers** or **rotamers**; an alkane has **infinitely many**
- the rotation is **not wholly free**, though: a **small barrier of 1-20 kJ mol^{-1}** stands in the way, owing to **weak repulsion between neighbouring bonds**, a repulsion called **torsional strain**

Conformations of ethane

- ethane (C_2H_6) has **one C–C bond, each carbon carrying three hydrogens**
- on a ball-and-stick model, **hold one carbon still and turn the other about the C–C axis**: the **hydrogens on one carbon take up endless positions relative to those on the other**, the **conformational isomers**, or conformers
- two cases stand at the **extremes**: hydrogens on the two carbons **as close as they can get** is the **eclipsed** conformation, **as far apart as they can get** the **staggered**; anything between is **skew**
- in every conformation the **bond angles and bond lengths stay the same**
- eclipsed and staggered forms are drawn as **Sawhorse** and as **Newman** projections

1 · Sawhorse projections

- the molecule is **viewed along its axis** and put on paper with the **central C–C bond as a longish straight line**, its **upper end leaning a little to right or left**
- the **front carbon sits at the lower end**, the **rear carbon at the upper**
- each carbon carries **three lines, one per hydrogen, 120° apart**; Figure 9.B shows ethane eclipsed and staggered this way

2 · Newman projections

- here the molecule is **viewed head-on down the C–C bond**
- the **carbon nearer the eye is a point**, its three hydrogens on **three lines 120° apart**
- the **rear carbon is a circle**, its three hydrogens on **shorter lines, again 120° apart**, attached to the rim; see Figure 9.C

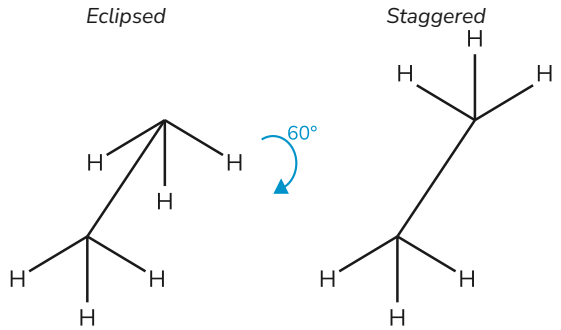


Figure 9.B Sawhorse projections of ethane

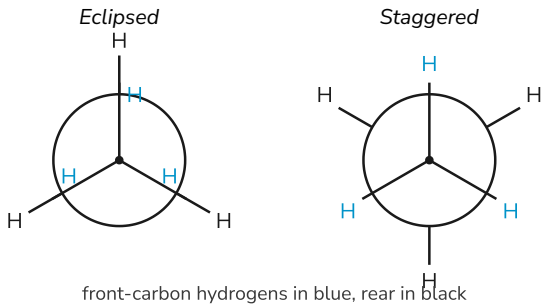


Figure 9.C Newman's projections of ethane

Relative stability of conformations

- in **staggered ethane the C–H electron clouds stand as far apart as they can: least repulsion, least energy, most stable**
- turn it towards the **eclipsed form and those clouds draw together**; the **repulsion grows**, the molecule **must carry more energy** to bear it, and is **less stable**
- that cloud-to-cloud repulsion, which sets a conformation's stability, is the **torsional strain**, and **how much of it there is depends on the angle turned about the C–C bond**, the **dihedral or torsional angle**
- of all ethane's conformations the **staggered has the least torsional strain, the eclipsed the most**; so **staggered is the more stable**, the one the molecule **mostly stays in**, its **preferred** conformation
- so **rotation about ethane's C–C bond is not wholly free**; still, the **gap between the two extremes is only about 12.5 kJ mol^{-1}**
- even at **ordinary temperatures, collisions give the molecule thermal energy enough to clear that 12.5 kJ mol^{-1}**, so for **practical purposes the rotation is almost free; no one has managed to separate and isolate ethane's conformers**

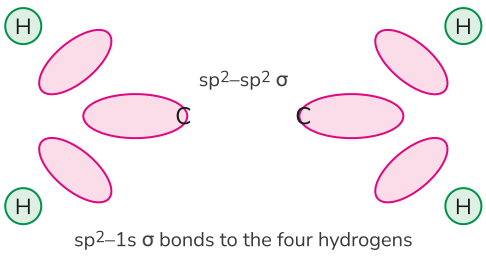


Figure 9.D Orbital picture of ethene depicting σ bonds only

B.1 Alkenes

- an alkene is an **unsaturated hydrocarbon with one double bond at least**
- its general formula: **one C=C between two carbons means two hydrogens fewer than the alkane**, so **C_nH_{2n}**
- alkenes are also **olefins**, oil-forming, since the first of them, **ethylene or ethene (C_2H_4)**, was found to **give an oily liquid with chlorine**

Structure of the double bond

- a C=C is **one strong σ bond** (enthalpy about **397 kJ mol^{-1}**), from **head-on overlap of *sp*^{2} orbitals**, plus **one weak π bond** (about **284 kJ mol^{-1}**) from **sideways overlap of the two carbons' 2*p* orbitals**
- it is **shorter, 134 pm, than a C–C single bond, 154 pm**
- the π bond is weak because the **sideways overlap of the 2*p* orbitals is poor**; so an alkene is a **store of loosely held, mobile electrons**, and **reagents hunting for electrons**, the **electrophilic reagents**, attack it readily
- that weak π bond leaves alkenes **less stable than alkanes**, and an **electrophile can turn one into a single-bonded compound**
- the **double bond as a whole, 681 kJ mol^{-1}**, is nonetheless **stronger than ethane's C–C, 348 kJ mol^{-1}**; Figure 9.D and Figure 9.E picture the orbitals of ethene

Nomenclature

- for an alkene's IUPAC name, take the **longest carbon chain that holds the double bond**
- **number it from the end nearer** — continued on sheet 13

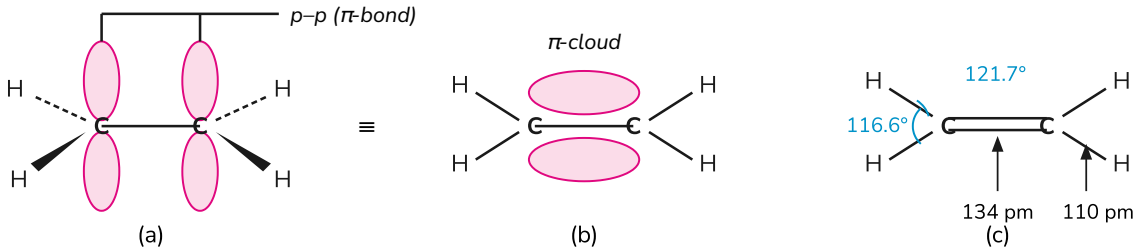


Figure 9.E Orbital picture of ethene showing formation of (a) π-bond, (b) π-cloud and (c) bond angles and bond lengths

- —the double bond, and the suffix **'ene' takes the place of 'ane'**
- the first member by formula, **CH_2** (n = 1 in C_nH_{2n}), is **methene**, but it **lives only an instant**; the **first stable alkene is C_2H_4, ethylene (common) or ethene (IUPAC)**

Structure	IUPAC name
$CH_3–CH=CH_2$	Propene
$CH_3–CH_2–CH=CH_2$	But-1-ene
$CH_3–CH=CH–CH_3$	But-2-ene
$CH_2=CH–CH=CH_2$	Buta-1,3-diene
$CH_2=C(CH_3)–CH_3$	2-Methylprop-1-ene
$CH_2=CH–CH(CH_3)–CH_3$	3-Methylbut-1-ene

Isomerism

- alkenes show **structural isomerism and geometrical isomerism** both
- **Structural isomerism** — as with the alkanes, **ethene (C_2H_4) and propene (C_3H_6) have one structure each**, but **above propene the structures multiply**; **C_4H_8** can be written **three ways**

$^1CH_2=^2CH–^3CH_2–^4CH_3$
I · But-1-ene (C_4H_8)
$^1CH_3–^2CH=^3CH–^4CH_3$
II · But-2-ene (C_4H_8)

Worked examples 9.g and 9.h

9.g Give IUPAC names to the four compounds below. **9.h** Then count the σ and π bonds in each.

(i)

$(CH_3)_2CH–CH=CH–CH_2–CH=CH–CH(C_2H_5)–CH_3$

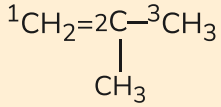
(ii)



(iii) $CH_2=C(CH_2CH_2CH_3)_2$

(iv) $CH_3–CH(CH_3)–CH=C((CH_2)_3CH_3)–CH_2–CH(C_2H_5)–CH_3$

- (i) **2,8-Dimethyl-3,6-decadiene** · σ 33, π 2
- (ii) **1,3,5,7-Octatetraene** · σ 17, π 4
- (iii) **2-*n*-Propylpent-1-ene** · σ 23, π 1
- (iv) **4-Ethyl-2,6-dimethyl-dec-4-ene** · σ 41, π 1



III · 2-Methylprop-1-ene (C_4H_8)

- **I and III**, and **II and III**, are **chain isomers**; **I and II** are **position isomers**

Worked example 9.i

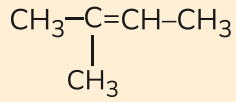
Draw and name every structural isomer of the alkene C_5H_{10}.

(a) $CH_2=CH–CH_2–CH_2–CH_3$

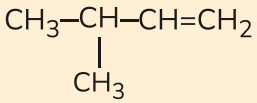
Pent-1-ene

(b) $CH_3–CH=CH–CH_2–CH_3$

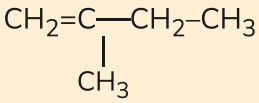
Pent-2-ene

(c) 

2-Methylbut-2-ene

(d) 

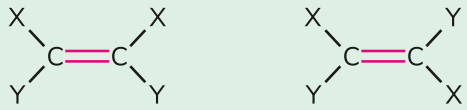
3-Methylbut-1-ene

(e) 

2-Methylbut-1-ene

Geometrical isomerism

- each carbon of a C=C has **two valences left**, filled by **two atoms or groups**; if the **two on each carbon differ**, the molecule is of the type **YXC=CXY**, and it can be set out in space **two ways**

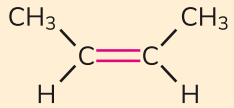


(a) *(b)*

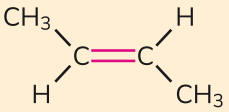
- in **(a) the like atoms, both X or both Y, sit on one side** of the double bond; in **(b) they sit across it**, on opposite sides
- so (a) and (b) **differ in geometry**, in **how the atoms or groups are placed in space**: they are **stereoisomers**
- they would be one and the same if the **groups could swing round the C=C**, but **rotation about a double bond is not free; it is restricted**
- to feel this, **nail two pieces of stiff cardboard together with two nails**, hold one and try to turn the other: **it will not turn**

- so the **restricted rotation about doubly bonded carbons** gives such compounds **different geometries**; stereoisomers of this kind are **geometrical isomers**
- type **(a)**, **like groups on the same side** of the double bond, is the ***cis*** isomer; type **(b)**, **like groups on opposite sides**, the ***trans***
- *cis* and *trans* isomers **share a structure but differ in configuration**, the arrangement of atoms or groups in space
- that difference shows in their **melting point, boiling point, dipole moment, solubility** and the like

- the **geometrical, or *cis-trans*, isomers of but-2-ene**:



cis-But-2-ene (b.p. 277 K)

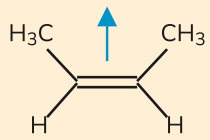


trans-But-2-ene (b.p. 274 K)

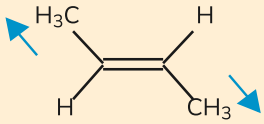
- the ***cis* form of an alkene is the more polar**: ***cis*-but-2-ene has a dipole moment of 0.33 Debye**, the ***trans* form next to none**, or, put another way

— continued on sheet 15

- —***trans*-but-2-ene is non-polar**; draw the two geometries and the reason is plain: in the ***trans* form the two methyls point opposite ways**, the **C–CH_3 bond dipoles cancel**, and the molecule ends up non-polar



cis-But-2-ene (μ = 0.33D)

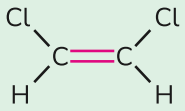


trans-But-2-ene (μ = 0)

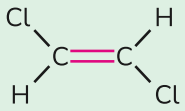
- among **solids the *trans* isomer melts higher than the *cis***
- *cis-trans* isomerism turns up as well in alkenes of the **XYC=CXZ** and the **XYC=CZW** type

Worked example 9.j

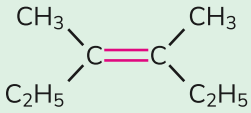
For (i) CHCl=CHCl and (ii) $C_2H_5CCH_3=CCH_3C_2H_5$, draw the *cis* and the *trans* isomer and name each.



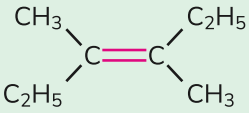
cis-1,2-Dichloroethene



trans-1,2-Dichloroethene



cis-3,4-Dimethylhex-3-ene



trans-3,4-Dimethylhex-3-ene

Worked example 9.k

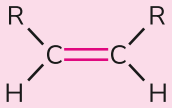
Which of these show *cis-trans* isomerism? (i) $(CH_3)_2C=CH–C_2H_5$; (ii) $CH_2=CBr_2$; (iii) $C_6H_5CH=CH–CH_3$; (iv) $CH_3CH=CClCH_3$

- **(iii) and (iv)**; in (i) and (ii) **one of the doubly bonded carbons carries two identical groups**

Preparation

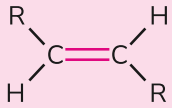
- **1 · From alkynes** — **partial reduction with a measured amount of dihydrogen** over **palladised charcoal partly deactivated by a poison**, sulphur compounds or quinoline, gives an alkene
- that partly deactivated palladised charcoal is **Lindlar's catalyst**, and the alkene it gives has ***cis* geometry**; **sodium in liquid ammonia**, by contrast, reduces an alkyne to the ***trans* alkene**

i) $R–C\equiv C–R + H_2 \xrightarrow{Pd/C}$



cis-Alkene

ii) $R–C\equiv C–R + H_2 \xrightarrow{Na/liq.\ NH_3}$



trans-Alkene

iii) $CH\equiv CH + H_2 \xrightarrow{Pd/C} CH_2=CH_2$

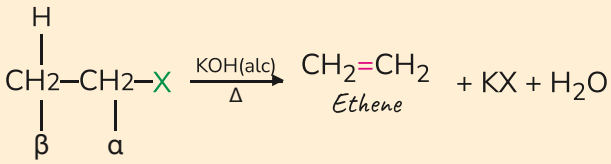
Ethyne → *Ethene*

iv) $CH_3–C\equiv CH + H_2 \xrightarrow{Pd/C} CH_3–CH=CH_2$

Propyne → *Propene*

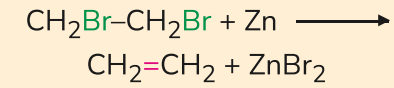
- will the **propene so made show geometrical isomerism**? think out the reason behind your answer
- **2 · From alkyl halides** — an alkyl halide (R–X) heated with **alcoholic potash**, that is potassium hydroxide dissolved in an alcohol, say — continued on sheet 16

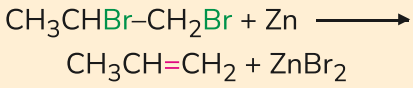
- —ethanol, **loses one molecule of halogen acid** and an alkene forms: **dehydrohalogenation**
- it is a **β-elimination**, the **hydrogen leaving from the β carbon**, the one next to the carbon that holds the halogen



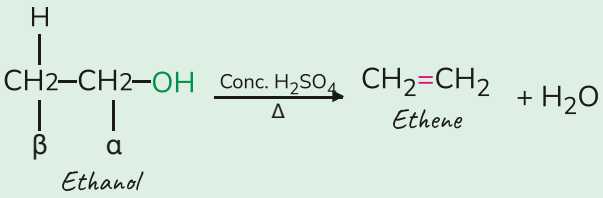
(X = Cl, Br, I)

- the **halogen and the alkyl group set the rate**: for halogens **iodine > bromine > chlorine**, for alkyl groups **tertiary > secondary > primary**
- **3 · From vicinal dihalides** — a dihalide with its **two halogens on adjacent carbons** is **vicinal**; **zinc metal strips a molecule of ZnX_2 from it**, leaving an alkene: **dehalogenation**





- **4 · From alcohols by acidic dehydration** — alcohols, as Unit 12 said in naming the homologous series, are the **hydroxy derivatives of alkanes, R–OH**, R being C_nH_{2n+1}
- **heated with concentrated sulphuric acid** an alcohol **sheds one molecule of water** and becomes an alkene; water lost from an alcohol under acid is **acidic dehydration**
- this too is a **β-elimination**: the **–OH carries off a hydrogen from the β-carbon**



Properties · physical

- as a class alkenes **look much like alkanes**, save in their **kinds of isomerism** and their **polarity**
- the **first three are gases, the next fourteen liquids, the rest solids**; **ethene, a colourless gas, smells faintly sweet**; the others have **neither colour nor smell**
- **water will not dissolve them**; **non-polar solvents**, benzene or petroleum ether, **dissolve them fairly well**
- boiling point **rises steadily with size, 20–30 K per $–CH_2$ added**; as with alkanes, a **straight chain boils higher than its branched isomer**

Properties · chemical

- alkenes are **rich in loosely held π electrons**, so they go in for **addition reactions**, an **electrophile adding across the C=C** to give the addition product
- some reagents add by a **free radical mechanism**; under special conditions an alkene can even undergo **free radical substitution**
- **oxidation and ozonolysis** are prominent too; the reactions run as follows

- **1 · Addition of dihydrogen** — an alkene takes up **one molecule of H_2 over finely divided nickel, palladium or platinum** and becomes an alkane (section A.2, Preparation)
- **2 · Addition of halogens** — **bromine or chlorine adds to give a vicinal dihalide**; **iodine does not add under normal conditions** — continued on sheet 17

- the **reddish orange of bromine in carbon tetrachloride fades** as the bromine adds at a site of unsaturation, which makes the reaction a **test for unsaturation**
- halogen addition is an **electrophilic addition**, going through a **cyclic halonium ion**, which higher classes take up

$CH_2{=}CH_2 + Br{-}Br \xrightarrow{CCl_4} BrCH_2{-}CH_2Br$

Ethene — *1,2-Dibromoethane*

$CH_3{-}CH{=}CH_2 + Cl{-}Cl \longrightarrow CH_3{-}CHCl{-}CH_2Cl$

Propene — *1,2-Dichloropropane*

- **3 · Addition of hydrogen halides — HCl, HBr and HI add to give alkyl halides**, reacting in the order **HI > HBr > HCl**; like halogen addition this is **electrophilic addition**, shown here with HBr on symmetrical and on unsymmetrical alkenes

HBr on a symmetrical alkene

- a **symmetrical alkene**, like groups on both carbons of the double bond, takes HBr by the **electrophilic addition mechanism**

$CH_2{=}CH_2 + H{-}Br \longrightarrow CH_3{-}CH_2{-}Br$

$CH_3{-}CH{=}CH{-}CH_3 + HBr \longrightarrow$

$CH_3{-}CH_2{-}\underset{\displaystyle Br}{\underset{|}{CH}}{-}CH_3$

HBr on an unsymmetrical alkene · Markovnikov rule

- how does H–Br add to **propene**? two products, **I and II**, are conceivable

$CH_3{-}CH{=}CH_2 + H{-}Br \longrightarrow$

$CH_3{-}CHBr{-}CH_3$ **I**

$CH_3{-}CH_2{-}CH_2Br$ **II**

- the Russian chemist **Markovnikov**, after a close study of such reactions, framed a generalisation in **1869**, the **Markovnikov rule**
- the rule: the **negative part of the addendum**, the adding molecule, **goes to the carbon that carries fewer hydrogens**
- so **product I, 2-bromopropane**, is expected, and in practice it **is the principal product**; the mechanism shows why

Mechanism

- hydrogen bromide supplies the **electrophile H^+**, which **attacks the double bond and makes a carbocation**

$CH_3{-}CH{=}CH_2 + H^+ \longrightarrow$

$CH_3{-}CH_2{-}{}^{+}CH_2$ (a) less stable primary carbocation

$CH_3{-}{}^{+}CH{-}CH_3$ (b) more stable secondary carbocation

- **(i)** the **secondary carbocation (b) is the more stable**, so it **forms faster and predominates**
- **(ii) Br^- then attacks (b)** and the product follows

$CH_3{-}C^{+}H{-}CH_3 + Br^{-} \longrightarrow CH_3{-}CHBr{-}CH_3$

2-Bromopropane (major product)

Anti Markovnikov addition · peroxide effect · Kharash effect

- **with a peroxide present, HBr adds to an unsymmetrical alkene such as propene the other way round**, against the Markovnikov rule
- this happens **with HBr alone, never with HCl or HI**
- **M.S. Kharash and F.R. Mayo** saw it in **1933 at the University of Chicago**; hence **peroxide effect**, **Kharash effect**, or **addition anti to Markovnikov rule**

$$CH_3–CH=CH_2 + HBr \xrightarrow{(C_6H_5CO)_2O_2} CH_3–CH_2–CH_2Br$$

1-Bromopropane

- the **secondary free radical of step (iii) is the more stable**, which is why **1-bromopropane is the major product**
- no peroxide effect with **HCl or HI**: the **H–Cl bond, 430.5 kJ mol^{-1}, is stronger than H–Br, 363.7 kJ mol^{-1}**, and a free radical **cannot break it**; the **H–I bond, 296.8 kJ mol^{-1}, is weaker**, but the **iodine radicals pair off into I_2** instead of adding to the double bond

Mechanism · a free radical chain

(i) $C_6H_5–CO–O–O–CO–C_6H_5 \xrightarrow{\text{Homolysis}}$

Benzoyl peroxide

$$2C_6H_5–CO–\dot{O}$$

$$2C_6H_5–CO–\dot{O} \longrightarrow 2\dot{C}_6H_5 + 2CO_2$$

(ii) $\dot{C}_6H_5 + H–Br \xrightarrow{\text{Homolysis}} C_6H_6 + \dot{B}r$

(iii) $CH_3–CH=CH_2 + \dot{B}r \xrightarrow{\text{Homolysis}}$

$$CH_3–\underset{\displaystyle Br}{\underset{|}{CH}}–\dot{C}H_2$$

(a) less stable primary free radical

$$CH_3–\dot{C}H–CH_2–Br$$

(b) more stable secondary free radical

(iv) $CH_3–\dot{C}H–CH_2Br + H–Br \xrightarrow{\text{Homolysis}}$

$$CH_3–CH_2–CH_2Br + \dot{B}r$$

major product

Worked example 9.1

HBr adds to hex-1-ene (i) with no peroxide about, and (ii) with peroxide present. Name each product by the IUPAC system.

(i) $CH_2=CH–CH_2–CH_2–CH_2–CH_3 + H–Br$

Hex-1-ene

$\xrightarrow{\text{No peroxide}}$

$$CH_3–CHBr–CH_2–CH_2–CH_2–CH_3$$

2-Bromohexane

(ii) $CH_2=CH–CH_2–CH_2–CH_2–CH_3 + H–Br$

$\xrightarrow{\text{Peroxide}}$

$$BrCH_2–CH_2–CH_2–CH_2–CH_2–CH_3$$

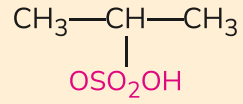
1-Bromohexane

- **4 · Adding sulphuric acid — concentrated H_2SO_4, cold, goes across the double bond Markovnikov fashion**, and an **alkyl hydrogen sulphate** results, by **electrophilic addition** — the equations follow on sheet 19

$$CH_2{=}CH_2 + H{-}OSO_3H \longrightarrow CH_3{-}CH_2{-}OSO_2{-}OH \text{ or } C_2H_5HSO_4$$

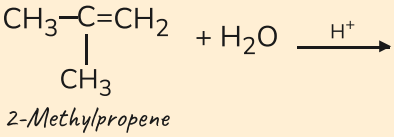
Ethyl hydrogen sulphate

$$CH_3{-}CH{=}CH_2 + HOSO_2OH \longrightarrow$$

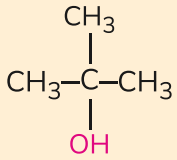


Propyl hydrogen sulphate

- **5 · Addition of water** — with a **few drops of concentrated sulphuric acid** present, an alkene **takes up water to give an alcohol**, again by the **Markovnikov rule**



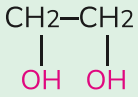
2-Methylpropene



2-Methylpropan-2-ol

- **6 · Oxidation** — **(a) cold, dilute, aqueous potassium permanganate, Baeyer's reagent**, turns an alkene into a **vicinal glycol**; the **$KMnO_4$ losing its colour is a test for unsaturation**

$$CH_2{=}CH_2 + H_2O + O \xrightarrow[273\ K]{\text{dil. } KMnO_4}$$



Ethane-1,2-diol (Glycol)

$$CH_3{-}CH{=}CH_2 + H_2O + O \xrightarrow[273\ K]{\text{dil. } KMnO_4} CH_3CH(OH)CH_2OH$$

Propane-1,2-diol

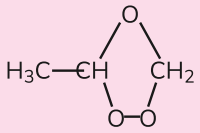
- **(b) acidic permanganate or acidic dichromate** oxidises an alkene to **ketones and/or acids**, depending on the alkene and the conditions

$$\underset{\text{2-Methylpropene}}{(CH_3)_2C{=}CH_2} \xrightarrow{KMnO_4/H^+} \underset{\text{Propan-2-one}}{(CH_3)_2C{=}O} + CO_2 + H_2O$$

$$\underset{\text{But-2-ene}}{CH_3{-}CH{=}CH{-}CH_3} \xrightarrow{KMnO_4/H^+} \underset{\text{Ethanoic acid}}{2CH_3COOH}$$

- **7 · Ozonolysis** — an **ozone molecule adds to the alkene to make an ozonide**, and **Zn–H_2O then splits the ozonide into smaller molecules**
- the reaction is a fine way of **locating a double bond** in an alkene or other unsaturated compound

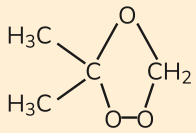
$$\underset{\text{Propene}}{CH_3CH{=}CH_2} + O_3 \longrightarrow$$



Propene ozonide

$$\xrightarrow{Zn + H_2O} \underset{\text{Ethanal}}{CH_3CHO} + \underset{\text{Methanal}}{HCHO}$$

$$\underset{\text{2-Methylpropene}}{(CH_3)_2C{=}CH_2} + O_3 \longrightarrow$$



Ozonide

$$\xrightarrow{Zn + H_2O} \underset{\text{Propan-2-one}}{(CH_3)_2C{=}O} + HCHO$$

- **8 · Polymerisation** — polythene bags and sheets are familiar; **polythene** is **a great many ethene molecules joined up at high temperature and pressure over a catalyst**
- the **huge molecules so made are polymers**, the reaction **polymerisation**, and the **simple compounds the polymers are built from are called** — continued on sheet 20

- —**monomers**; other alkenes polymerise too

$$n(CH_2{=}CH_2) \xrightarrow[\text{Catalyst}]{\text{High temp./pressure}} —(CH_2{-}CH_2)—_n$$

Polythene

$$n(CH_3{-}CH{=}CH_2) \xrightarrow[\text{Catalyst}]{\text{High temp./pressure}} —(CH(CH_3){-}CH_2)—_n$$

Polypropene

- from polymers come **plastic bags and squeeze bottles, refrigerator dishes and toys, pipes, and the cabinets of radios and T.V. sets**; **polypropene** is moulded into **milk crates, plastic buckets and the like**
- common as they now are, the **heavy use of polythene and polypropylene worries us all**

C.1 Alkynes

- alkynes, like alkenes, are **unsaturated**: they hold **one C≡C at least**
- they carry **fewer hydrogens still than alkenes or alkanes**; general formula $\mathbf{C_nH_{2n-2}}$
- the **first stable alkyne is ethyne, better known as acetylene**; mixed with oxygen it gives the **oxyacetylene flame used in arc welding**
- a **great many organic compounds start from alkynes**, which makes the class worth studying

Nomenclature and isomerism

- the **common system names alkynes as derivatives of acetylene**; the **IUPAC system takes the alkane's name and swaps 'ane' for 'yne'**, the **first triply bonded carbon giving the triple bond's position**; see Table 9.C

Table 9.C Common and IUPAC Names of Alkynes (C_nH_{2n-2})

Value of n	Formula	Structure	Common name	IUPAC name
2	C_2H_2	H–C≡CH	Acetylene	Ethyne
3	C_3H_4	CH_3–C≡CH	Methylacetylene	Propyne
4	C_4H_6	CH_3CH_2–C≡CH	Ethylacetylene	But-1-yne
4	C_4H_6	CH_3–C≡C–CH_3	Dimethylacetylene	But-2-yne

- **ethyne and propyne have one structure each**; butyne has two, **(i) but-1-yne and (ii) but-2-yne**, which differ in **where the triple bond sits**: **position isomers**
- the next homologue, $\mathbf{C_5H_8}$: set five carbons in a **straight chain**, or with a **side chain**, and these structures come out

$$H^1C{\equiv}^2C{-}^3CH_2{-}^4CH_2{-}^5CH_3$$

I · Pent-1-yne

$$H_3{}^1C{-}^2C{\equiv}^3C{-}^4CH_2{-}^5CH_3$$

II · Pent-2-yne

$$H_3{}^4C{-}_3CH(CH_3){-}^2C{\equiv}^1CH$$

III · 3-Methylbut-1-yne

- **I and II are position isomers**; **I and III**, and **II and III**, are **chain isomers**

Worked example 9.m

Draw and name every isomer of the fifth alkyne in the series, and say what kind of isomerism each pair shows.

- the fifth alkyne is $\mathbf{C_6H_{10}}$; its isomers are set out on sheet 21

Worked example 9.m · the isomers of C_6H_{10}

(a) $HC{\equiv}C{-}CH_2{-}CH_2{-}CH_2{-}CH_3$
Hex-1-yne

(b) $CH_3{-}C{\equiv}C{-}CH_2{-}CH_2{-}CH_3$
Hex-2-yne

(c) $CH_3{-}CH_2{-}C{\equiv}C{-}CH_2{-}CH_3$
Hex-3-yne

(d) $HC{\equiv}C{-}CH(CH_3){-}CH_2{-}CH_3$
3-Methylpent-1-yne

(e) $HC{\equiv}C{-}CH_2{-}CH(CH_3){-}CH_3$
4-Methylpent-1-yne

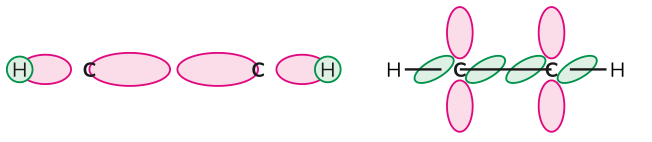
(f) $CH_3{-}C{\equiv}C{-}CH(CH_3){-}CH_3$
4-Methylpent-2-yne

(g) $HC{\equiv}C{-}C(CH_3)_2{-}CH_3$
3,3-Dimethylbut-1-yne

- the pairs show **position isomerism** or **chain isomerism**, according as the triple bond moves or the skeleton changes

Structure of the triple bond

- **ethyne** is the simplest alkyne; Figure 9.F shows its structure
- each carbon has **two *sp* hybrid orbitals**; the **C–C σ bond comes of two of them overlapping head-on**, and the **other *sp* orbital on each carbon overlaps along the axis with a hydrogen 1*s***, giving **two C–H σ bonds**
- the **H–C–C angle is 180°**
- each carbon keeps **two unhybridised *p* orbitals, at right angles to each other and to the C–C σ bond**; the 2*p* orbitals of one carbon **lie parallel to those of the other**, and **sideways overlap gives two π bonds**
- so ethyne holds **one C–C σ, two C–H σ and two C–C π bonds**
- the **C≡C bond, 823 kJ mol^{-1}, is stronger than C=C, 681, and C–C, 348 kJ mol^{-1}**, and **shorter, 120 pm, than C=C, 133 pm, and C–C, 154 pm**
- the **electron cloud between the carbons is cylindrically symmetrical about the axis**, so **ethyne is linear**



sp–sp and sp–1s σ overlaps (a) two p–p π overlaps, at right angles (b)

Figure 9.F Orbital picture of ethyne showing (a) sigma overlaps (b) pi overlaps.

Preparation

- **1 · From calcium carbide — industry makes ethyne by treating calcium carbide with water**
- the carbide comes from **quick lime heated with coke**, and the quick lime from **heating limestone**:

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$CaO + 3C \longrightarrow \underset{\text{Calcium carbide}}{CaC_2} + CO$$

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$$

- **2 · From vicinal dihalides** — **alcoholic potassium hydroxide dehydrohalogenates a vicinal dihalide**: **one molecule of hydrogen halide leaves** and an **alkenyl halide** forms, which **sodamide** then takes on to the **alkyne**

$$CH_2Br{-}CH_2Br \xrightarrow[-HBr]{\text{alc. KOH}} CH_2{=}CHBr \xrightarrow[-HBr]{NaNH_2} HC{\equiv}CH$$

Properties · physical

- alkynes follow the **same trend as alkenes and alkanes**: the **first three are gases, the next eight liquids, the rest solids**
- all are **colourless**; **ethyne has a characteristic odour**, the others none
- **weakly polar**, **lighter than water and immiscible with it**, but **soluble in organic solvents** such as ethers, carbon tetrachloride and benzene
- **melting point, boiling point and density all rise with molar mass**

Properties · chemical

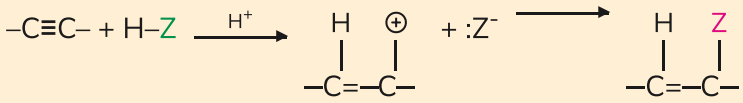
- alkynes show an **acidic character**, **addition reactions** and **polymerisation**
- **A · Acidic character** — **sodium metal, and sodamide ($NaNH_2$), are both strong bases**; with ethyne they give **sodium acetylide and dihydrogen**
- **ethene and ethane do nothing of the kind**, so **ethyne is acidic by comparison**; why? the answer lies in **structure and hybridisation**
- ethyne's hydrogens sit on ***sp* carbons**, ethene's on sp^2, ethane's on sp^3
- with the **most *s* character, 50%**, the ***sp* orbital is the most electronegative**, and **pulls the C–H electron pair towards carbon harder** than sp^2 in ethene or sp^3 in ethane does
- so **ethyne gives up its hydrogens as protons more readily**: a **hydrogen on a triply bonded carbon is acidic**, though **not every hydrogen of an alkyne is**

$$HC{\equiv}CH + Na \longrightarrow \underset{\text{Monosodium ethynide}}{HC{\equiv}C^-Na^+} + 1/2\ H_2$$

$$HC{\equiv}C^-Na^+ + Na \longrightarrow \underset{\text{Disodium ethynide}}{Na^+C^-{\equiv}C^-Na^+} + 1/2\ H_2$$

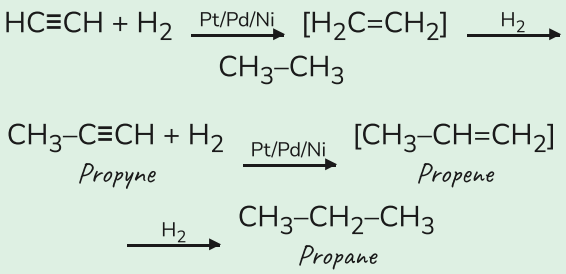
$$CH_3{-}C{\equiv}C{-}H + Na^+NH_2^- \longrightarrow \underset{\text{Sodium propynide}}{CH_3{-}C{\equiv}C^-Na^+} + NH_3$$

- **alkenes and alkanes give none of these reactions**, which makes them a way of **telling an alkyne from an alkene or alkane**; what would **but-1-yne and but-2-yne** do?
- acidity runs: **i) $CH{\equiv}CH > H_2C{=}CH_2 > CH_3{-}CH_3$**
- **ii) $HC{\equiv}CH > CH_3{-}C{\equiv}CH >> CH_3{-}C{\equiv}C{-}CH_3$**
- **B · Addition reactions** — with a triple bond to spend, an alkyne **adds two molecules** of dihydrogen, halogen, hydrogen halide and so on; the addition product forms **in the steps set out on sheet 23**

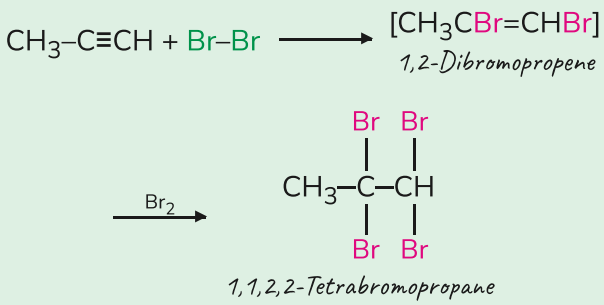


- the charged intermediate is the **vinylic cation**, and which product forms turns on **its stability**; on an **unsymmetrical alkyne the addition follows the Markovnikov rule**, and **most alkyne reactions are electrophilic additions**; a few follow

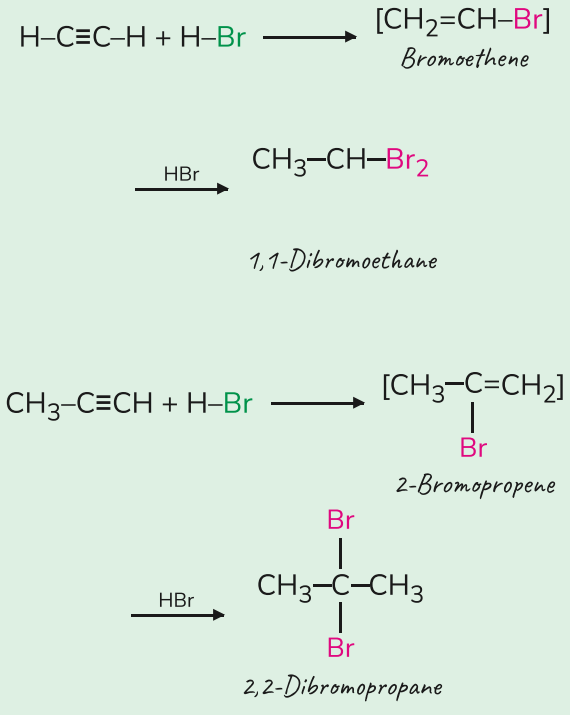
- **(i) Addition of dihydrogen**



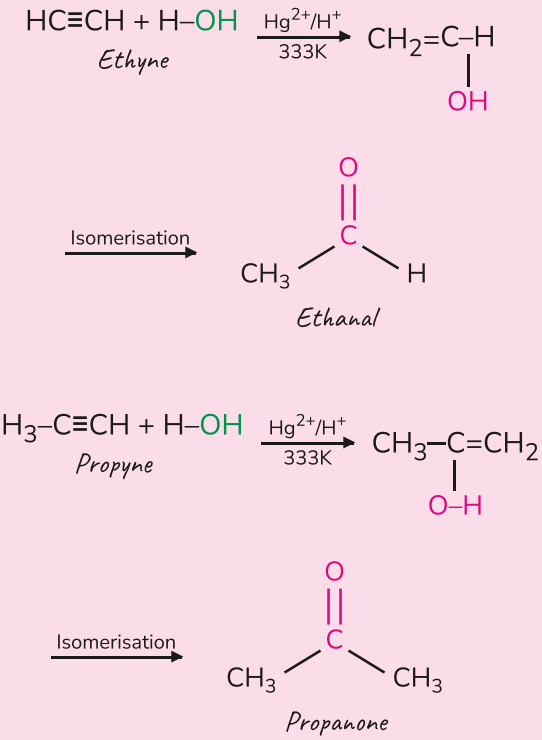
- **(ii) Addition of halogens**



- the **reddish orange of bromine in carbon tetrachloride is discharged**: a **test for unsaturation**
- **(iii) Adding a hydrogen halide — HCl, HBr or HI adds twice over**, giving a ***gem* dihalide, both halogens on one carbon**

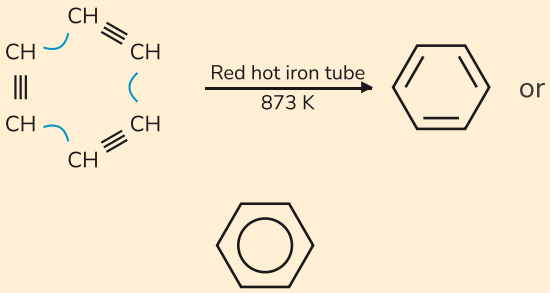


- **(iv) Addition of water** — alkynes, like alkanes and alkenes, **neither mix with water nor react with it**
- yet at **333 K, warmed with dilute sulphuric acid and mercuric sulphate**, an alkyne **takes up one molecule of water** and a **carbonyl compound** results



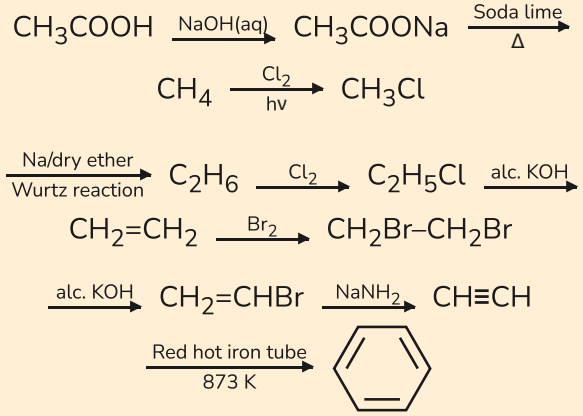
- **(v) Polymerisation — (a) Linear polymerisation**: in the right conditions **ethyne strings itself into polyacetylene, or polyethyne**, a **high molecular weight polyene** built of repeating **(CH=CH–CH=CH)** units, written **—$(CH=CH–CH=CH)_n$—**
- given the right conditions the polymer **carries an electric current**

- a **thin film of polyacetylene serves as a battery electrode**: such films **conduct well**, and are **lighter and cheaper than metal conductors**
- **(b) Cyclic polymerisation — ethyne passed through a red hot iron tube at 873K** polymerises into a ring: **three molecules make one benzene**
- benzene is the **starting point for its own derivatives, for dyes, drugs and a host of other organic compounds**, and this is the **best road from aliphatic to aromatic**



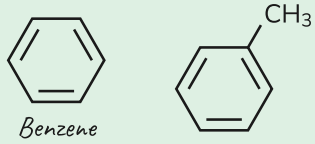
Worked example 9.n

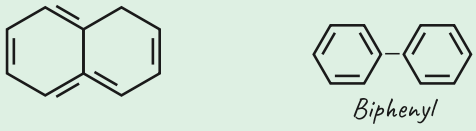
Set out a route from ethanoic acid to benzene.



D.1 Aromatic hydrocarbons

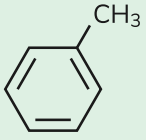
- these are the **arenes**; most **smell pleasant** (Greek *aroma*, pleasant smelling), whence **aromatic compounds**
- most were found to **hold a benzene ring**, which is **highly unsaturated**, yet in **most reactions of aromatic compounds the ring keeps its unsaturation**
- some aromatic hydrocarbons **have no benzene ring** but **another highly unsaturated ring** instead: those with a benzene ring are **benzenoids**, the rest **non-benzenoids**

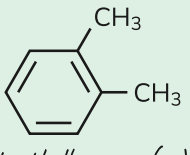


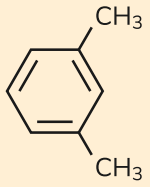


Nomenclature and isomerism

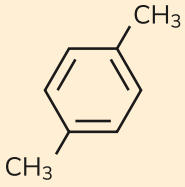
- Unit 8 covered the naming and isomerism of arenes; **all six hydrogens of benzene are equivalent**, so it gives **one monosubstituted product only**
- replace **two hydrogens**, by like or unlike monovalent atoms or groups, and **three position isomers** arise: **1,2 or 1,6 is *ortho* (o–), 1,3 or 1,5 *meta* (m–), 1,4 *para* (p–)**







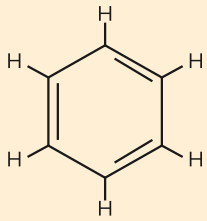
1,3-Dimethylbenzene (m-Xylene)



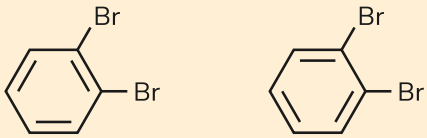
1,4-Dimethylbenzene (p-Xylene)

Structure of benzene

- **Michael Faraday isolated benzene in 1825**; its formula **C_6H_6** speaks of **heavy unsaturation**, yet **fits no pattern** set by the alkanes, alkenes and alkynes met earlier in this unit; what might its structure be?
- its **singular properties and unusual stability** kept the question open for **years**
- benzene proved **stable**, and **formed a triozonide**, which points to **three double bonds**; it gave **one monosubstituted derivative only**, so **all six carbons and all six hydrogens are alike**
- on that evidence **August Kekulé in 1865** proposed a **ring of six carbons, single and double bonds alternating, one hydrogen on each carbon**



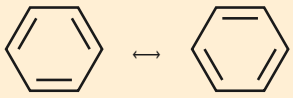
- the Kekulé structure allows **two isomeric 1,2-dibromobenzenes**: in one the **bromines sit on doubly bonded carbons**, in the other on **singly bonded** ones



Friedrich August Kekulé (7th September 1829–13th July 1896)

- a **German chemist, born 1829 at Darmsdt**; **Professor in 1856, Fellow of the Royal Society in 1875**
- his great contribution to structural organic chemistry came in two parts: in **1858 the proposal that carbon atoms join one another into chains**, and in **1865 the answer to the puzzle of benzene: such a chain can close into a ring**
- his **dynamic structural formula for benzene** underlies its **modern electronic structure**
- he later told how the structure came to him (1890): he had been writing at his textbook, but the work would not move and his thoughts were elsewhere; he swung his chair round to the fire and nodded off
- once more atoms danced before his eyes; the small groups stayed modestly back, while his inner eye, sharpened by many such visions, picked out large structures of many shapes: long rows, some packed closer than others, writhing and turning like snakes
- then a snake caught hold of its tail, and the shape spun mockingly in front of him; he woke as if struck by lightning, and the remainder of the night went on drawing out what the hypothesis meant
- his moral: **learn to dream, and perhaps learn the truth, but never make a dream public before the waking mind has approved it**
- a hundred years on, at his centenary, a group of **polybenzenoid compounds were named Kekulenes** in his honour

- yet benzene gives **one ortho disubstituted product only**; Kekulé met the difficulty by proposing that the **double bonds oscillate**:



- even so amended, the Kekulé structure **cannot account for benzene's unusual stability**, nor for its **preference for substitution over addition**; **resonance** later did

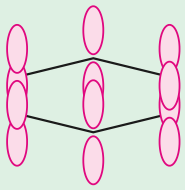
Resonance and stability of benzene

- in **Valence Bond Theory** the oscillating double bonds become **resonance**: benzene is a **hybrid of several resonating structures**, Kekulé's **A and B the main contributors**
- the hybrid is drawn as a **hexagon with a circle, or a dotted circle, inside**, as in **(C)**; the **circle stands for the six electrons delocalised over the six ring carbons**



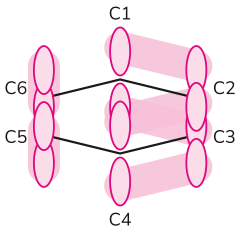
The orbital picture

- orbital overlap gives a **clearer picture**; **all six carbons are sp^2**
- **two sp^2 orbitals on each carbon overlap with those of its neighbours**, making **six C—C σ bonds, all in the hexagonal plane**
- the **third sp^2 orbital overlaps a hydrogen *s* orbital**, giving **six C—H σ bonds**
- that leaves each carbon **one unhybridised *p* orbital, standing at right angles to the ring**:

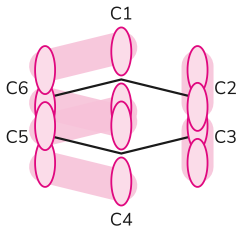


σ framework, one p orbital on each carbon

- the *p* orbitals stand **close enough to overlap sideways into π bonds**, and there are **two equally good ways of pairing them into three π bonds**: **C_1–C_2, C_3–C_4, C_5–C_6** or **C_2–C_3, C_4–C_5, C_6–C_1**, as the figures show



(a)



(b)

Figure 9.G (a)

- Figure 9.G (a) and (b) answer to the **two Kekulé structures with their π bonds localised**; the internuclear distance — continued on sheet 27

- —between every pair of ring carbons is, by **X-ray diffraction, the same**; so each carbon's *p* orbital is **as likely to overlap with one neighbour as with the other** (Figure 9.G (c))
- the result is drawn as **two doughnuts of electron cloud, one over the ring plane and one under it** (Figure 9.G (d))
- the **six π electrons are delocalised**, free to move about **all six carbon nuclei** rather than any two as in (a) or (b)
- a **delocalised π cloud is held more tightly by the carbon nuclei than one confined between two carbons**, so **benzene outdoes the hypothetical cyclohexatriene in stability**
- X-ray data also show benzene **planar**; were A or B alone right there would be **two C—C bond lengths**, yet **all six are alike, 139 pm, between a C—C single bond, 154 pm, and a double, 133 pm**
- with **no true double bond**, benzene is **loath to undergo addition under ordinary conditions**, which accounts for its odd behaviour

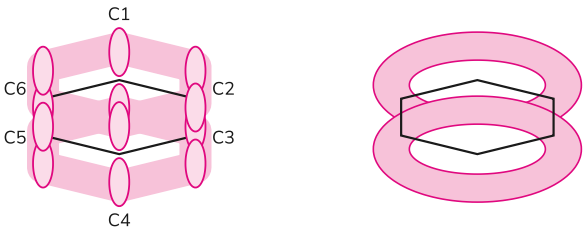


Figure 9.G (c) **(d)** (electron cloud)

Aromaticity

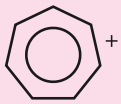
- benzene was the **parent 'aromatic' compound**; the name now covers **any ring system, benzene ring or not**, that has
- **(i) planarity**
- **(ii) full delocalisation of the π electrons round the ring**
- **(iii) (4n + 2) π electrons in the ring**, n an integer (n = 0, 1, 2, . . .)
- this is **Hückel's Rule**

Some aromatic compounds



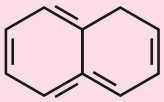
Benzene

Cyclopentadienyl anion

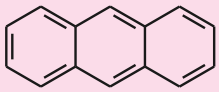


Cycloheptatrienyl cation

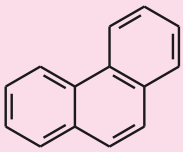
(n = 1, 6π electrons)



Naphthalene (n = 2, 10 π electrons)



Anthracene

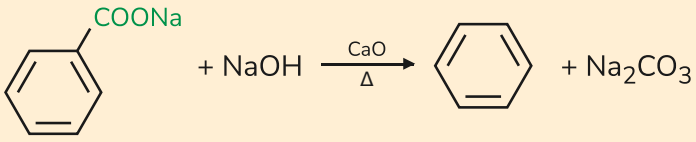


Phenanthrene

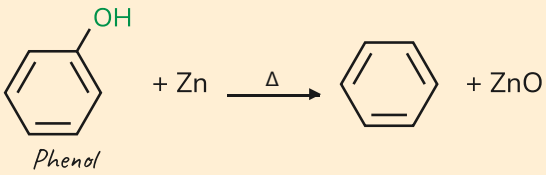
(n = 3, 14 π electrons)

Preparation of benzene

- commercially benzene is **taken from coal tar**; in the laboratory these methods serve
- **(i) Cyclic polymerisation of ethyne** (section C.1, Properties)
- **(ii) Decarboxylation of aromatic acids** — the **sodium salt of benzoic acid, heated with sodalime**, gives benzene



- **(iii) Reduction of phenol** — **phenol vapour over heated zinc dust** comes out as benzene



Phenol

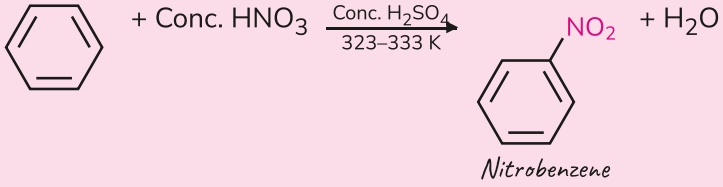
Properties · physical

- aromatic hydrocarbons are **non-polar**, mostly **colourless liquids or solids with a characteristic aroma**
- **naphthalene balls**, familiar from toilets and wardrobes, owe their use to the compound's **distinctive smell and moth-repellent property**
- arenes **do not mix with water** but **mix readily with organic solvents**, and they **burn with a sooty flame**

Properties · chemical

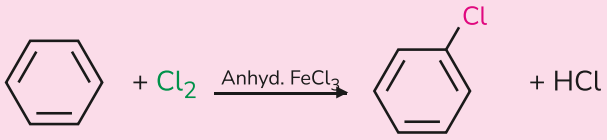
- what marks an arene is **electrophilic substitution**; under special conditions **addition and oxidation** happen too
- the usual ones: **halogenation and nitration, sulphonation, and the Friedel Craft's alkylation and acylation**, the attacking reagent in each an **electrophile (E^+)**
- **(i) Nitration** — benzene **heated with concentrated nitric acid plus concentrated sulphuric acid, the nitrating mixture**, takes a **nitro group onto the ring**
- **(ii) Halogenation** — with a **Lewis acid**, anhydrous **$FeCl_3$, $FeBr_3$ or $AlCl_3$**, an arene and a halogen give a **haloarene**
- **(iii) Sulphonation** — putting a **sulphonic acid group in place of a ring hydrogen**; benzene is **heated with fuming sulphuric acid, oleum**

(i)



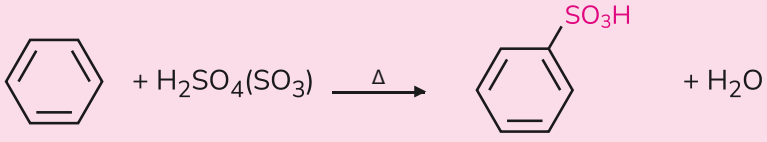
Nitrobenzene

(ii)



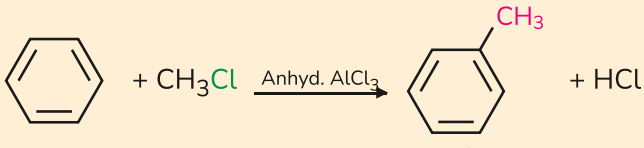
Chlorobenzene

(iii)

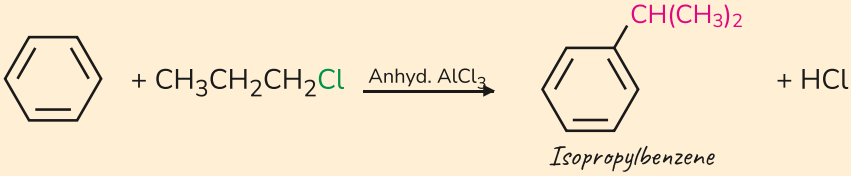


Benzenesulphonic acid

- **(iv) Friedel-Crafts alkylation** — benzene and an **alkyl halide over anhydrous aluminium chloride** give an **alkylbenzene**

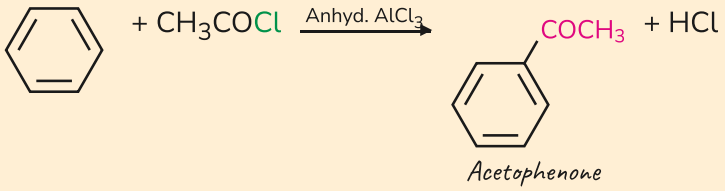


Toluene

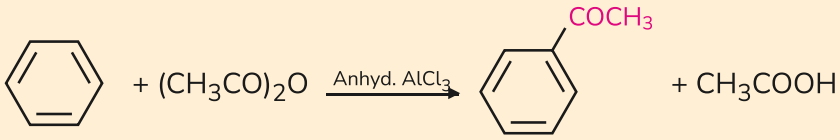


Isopropylbenzene

- why does **1-chloropropane give isopropyl benzene rather than *n*-propyl benzene**?
- **(v) Friedel-Crafts acylation** — benzene with an **acyl halide or an acid anhydride**, a **Lewis acid ($AlCl_3$)** present, gives an **acyl benzene**; the equations follow on sheet 29

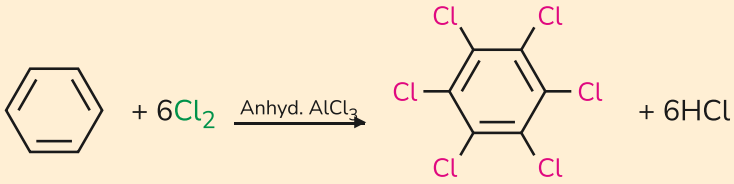


Acetophenone



Acetophenone

- given an **excess of the electrophilic reagent**, substitution can **go on**, the electrophile **taking the ring's other hydrogens one after another**; benzene with **excess chlorine over anhydrous $AlCl_3$** goes all the way to **hexachlorobenzene (C_6Cl_6)**



Hexachlorobenzene

Mechanism of electrophilic substitution

- the evidence has S_E reactions (S substitution, E electrophilic) running in **three steps**:
- **(a)** the **electrophile is generated**
- **(b)** a **carbocation intermediate forms**
- **(c)** a **proton leaves** that intermediate
- **(a) Generation of the electrophile $E^{\oplus}$** — in chlorination, alkylation and acylation, **anhydrous $AlCl_3$, a Lewis acid, combines with the reagent** and sets free **$Cl^{\oplus}$, $R^{\oplus}$ or $RC^{\oplus}O$, the acylium ion**

$$Cl{-}Cl + AlCl_3 \longrightarrow Cl^+ + AlCl_4^-$$

$$R{-}Cl + AlCl_3 \longrightarrow R^+ + AlCl_4^-$$

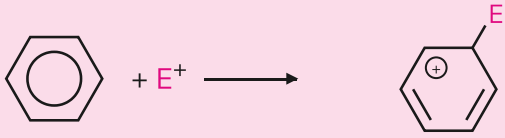
$$RCO{-}Cl + AlCl_3 \longrightarrow \underset{\text{acylium ion}}{RC^+O} + AlCl_4^-$$

- in nitration the electrophile is the **nitronium ion, $NO_2^{\oplus}$**, made when **sulphuric acid hands a proton to nitric acid**:

Step I $HNO_3 + H_2SO_4 \rightleftharpoons \underset{\text{Protonated nitric acid}}{H_2O^+{-}NO_2} + HSO_4^-$

Step II $H_2O^{+}{-}NO_2 \rightleftharpoons \underset{\text{Nitronium ion}}{NO_2^+} + H_2O$

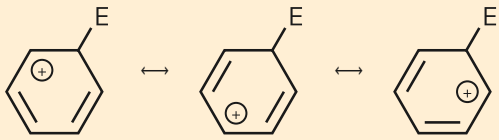
- notice that here **sulphuric acid plays the acid and nitric acid the base**: a plain **acid-base equilibrium**
- **(b) Formation of the carbocation, the arenium ion** — the electrophile's attack gives a **σ-complex, or arenium ion**, in which **one carbon has gone sp^3**



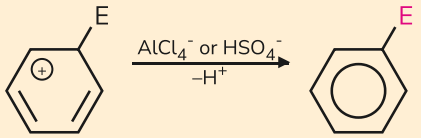
sigma complex (arenium ion)

- the arenium ion is **steadied by resonance**, drawn out on sheet 30

- the **arenium ion's resonance structures**, the positive charge moving round the ring:

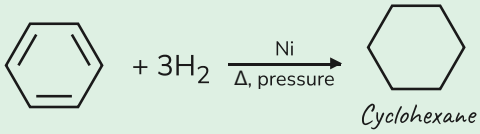


- the σ complex, or arenium ion, has **lost its aromatic character**: **delocalisation halts at the sp^3 carbon**
- **(c) Removal of proton** — to **win the aromatic character back**, the σ-complex **gives up a proton from the sp^3 carbon** when **$[AlCl_4]^-$** attacks (halogenation, alkylation, acylation) or **$[HSO_4]^-$** does (nitration)



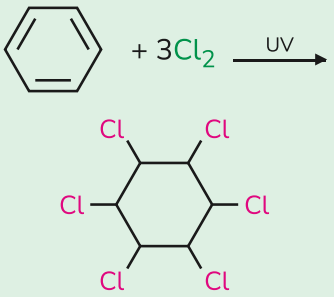
Addition reactions

- under **forcing conditions, high temperature and/or pressure over nickel**, benzene **hydrogenates to cyclohexane**



Cyclohexane

- in **ultra-violet light three chlorine molecules add**, giving **benzene hexachloride, $C_6H_6Cl_6$**, also called **gammaxane**



Benzene hexachloride (BHC)

- **Combustion** — heated in air, benzene **burns with a sooty flame** to **CO_2 and H_2O**

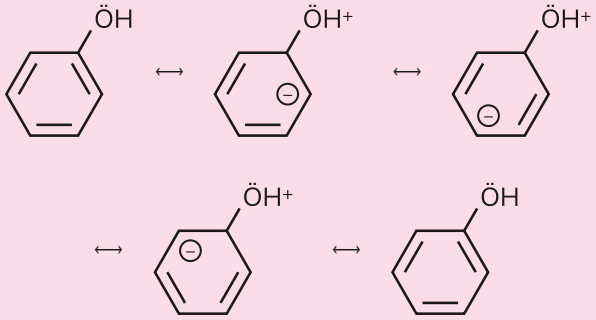
$$C_6H_6 + 15/2\ O_2 \longrightarrow 6CO_2 + 3H_2O$$

- and for **any hydrocarbon** the combustion runs

$$C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$$

Directive influence of a group already on the ring

- substitute a **monosubstituted benzene** a second time and the **three disubstituted products do not come in equal amounts**
- two patterns are seen: **mainly ortho and para**, or **mainly meta**
- which pattern appears depends on the **substituent already on the ring, not on the group coming in**: the **directive influence** of substituents; the reasons follow
- **Ortho and para directing groups** — groups that **send the newcomer to the ortho and para positions**; the **phenolic –OH** is the example, phenol being a **resonance hybrid** of the structures below

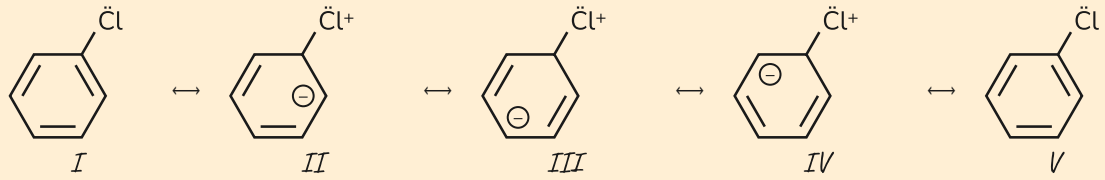


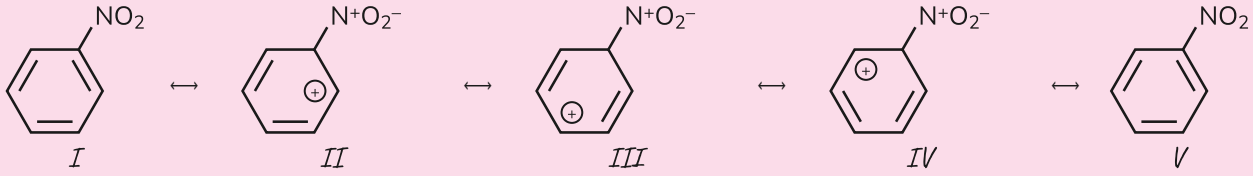
- the structures show **electron density piling up at the o– and p– positions**, so **substitution goes mainly there**
- the **–I effect of –OH** works too, **thinning the ortho and para electron density a little**, but **resonance raises it more than –I lowers it**; so **–OH activates the ring towards attack by** — continued on sheet 31

- —an electrophile; **–NH_2, –NHR, –$NHCOCH_3$, –OCH_3, –CH_3, –C_2H_5** and the like are **activating** too
- in an **aryl halide the halogen deactivates moderately**: its **strong –I effect drains electron density from the ring** and makes **further substitution harder**
- yet by **resonance the o– and p– positions still hold more electron density than the m-**, so **halogens are o– and p– directing as well**; chlorobenzene's resonance structures are drawn below

Meta directing groups

- groups that **send the newcomer to the meta position**: **–NO_2, –CN, –CHO, –COR, –COOH, –COOR, –SO_3H** and so on
- take the **nitro group**: its **strong –I effect cuts the ring's electron density**, and nitrobenzene is a **resonance hybrid** of the structures below



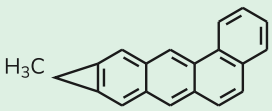


- here the **ring as a whole loses electron density**, so substitution is harder and such groups are **'deactivating groups'**; the **o– and p– positions are poorer in electrons than the meta**, so the **electrophile goes for the comparatively electron rich meta position**: **meta substitution**

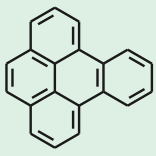
D.2 Carcinogenicity and toxicity

- **benzene**, and the **polynuclear hydrocarbons of more than two fused benzene rings**, are **toxic** and held to be **carcinogenic**, cancer producing
- such polynuclear hydrocarbons arise from the **incomplete combustion of organic matter**: **tobacco, coal, petroleum**; once in the body they go through **a chain of biochemical reactions** and end by **damaging DNA and causing cancer**; some are drawn below

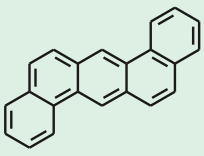
Some carcinogenic hydrocarbons



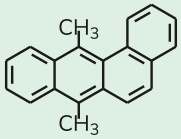
3-Methylcholanthrene



1,2-Benzpyrene



1,2,5,6-Dibenzanthracene



9,10-Dimethyl-1,2-benzanthracene

Summary

- a **hydrocarbon holds carbon and hydrogen only**; hydrocarbons come chiefly from **coal and petroleum**, our great **sources of energy**
- **petrochemicals** are the leading raw materials for a **vast range of commercial products**
- **liquefied petroleum gas, LPG, and compressed natural gas, CNG**, the mainstays of **household fuel and of motoring**, come from petroleum
- by structure, hydrocarbons are **open chain saturated (alkanes)** or **unsaturated (alkenes, alkynes)**, **cyclic (alicyclic)**, or **aromatic**
- alkanes: **free radical substitution, combustion, oxidation, aromatization**; alkenes and alkynes: **addition, mostly electrophilic**; arenes, for all their unsaturation, **electrophilic substitution** for the most part, with **addition reserved for special conditions**
- in alkanes there is **conformational isomerism**, the **C–C σ bonds turning freely**; of ethane's **staggered and eclipsed** forms the **staggered is the more stable, its hydrogens furthest apart**
- alkenes show **geometrical (*cis-trans*) isomerism**, since **rotation about C=C is restricted**
- **benzene and the benzenoids are aromatic**; **aromaticity** belongs to compounds with the electronic structure the **Hückel (4n+2)π electron rule** describes
- the **groups on a benzene ring decide whether it is activated or deactivated** towards further electrophilic substitution, and **where the newcomer goes**
- some **polynuclear hydrocarbons with fused benzene rings are carcinogenic**

Exercises

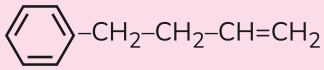
E1 Ethane turns up when methane is chlorinated. How does that come about?

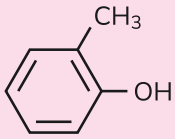
E2 Give each compound its IUPAC name:

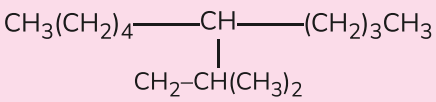
(a) $CH_3CH=C(CH_3)_2$

(b) $CH_2=CH–C\equiv C–CH_3$

(c) 

(d) 

(e) 

(f) 

(g) $CH_3–CH=CH–CH_2–CH=CH–\underset{\displaystyle C_2H_5}{\underset{|}{CH}}–CH_2–CH=CH_2$

E3 Draw every isomer with the stated number of multiple bonds, and name each: (a) C_4H_8, one double bond; (b) C_5H_8, one triple bond.

E4 Name, by the IUPAC system, what ozonolysis gives from (i) Pent-2-ene; (ii) 3,4-Dimethylhept-3-ene; (iii) 2-Ethylbut-1-ene; (iv) 1-Phenylbut-1-ene.

Exercises

E5 Ozonolysis of an alkene 'A' yields ethanal together with pentan-3-one. Draw 'A' and give its IUPAC name.

E6 An alkene 'A' has three C–C σ bonds, eight C–H σ bonds and one C–C π bond. Ozonolysis of 'A' gives two moles of an aldehyde of molar mass 44 u. What is the IUPAC name of 'A'?

E7 Ozonolysis of an alkene gives propanal and pentan-3-one. Give the alkene's structural formula.

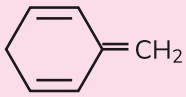
E8 Write the combustion equation for (i) Butane; (ii) Pentene; (iii) Hexyne; (iv) Toluene.

E9 Draw hex-2-ene as its *cis* and its *trans* isomer. Which boils higher, and why?

E10 Three double bonds and yet extraordinarily stable: how is benzene's stability accounted for?

E11 What must be true of a system for it to be aromatic?

E12 Why is none of the following aromatic?

(i)  (ii)  (iii) 

E13 Starting from benzene, how would you make (i) *p*-nitrobromobenzene; (ii) *m*-nitrochlorobenzene; (iii) *p*-nitrotoluene; (iv) acetophenone?

E14 In $H_3C–CH_2–C(CH_3)_2–CH_2–CH(CH_3)_2$, pick out the 1°, 2° and 3° carbons and say how many hydrogens each carries.

E15 How does branching the chain of an alkane change its boiling point?

E16 HBr on propene gives 2-bromopropane, but with benzoyl peroxide present it gives 1-bromopropane. Account for this, with mechanisms.

E17 What does ozonolysis of 1,2-dimethylbenzene (*o*-xylene) give? How does the outcome bear out Kekulé's structure for benzene?

E18 Put benzene, *n*-hexane and ethyne in order of falling acidity, and explain the order.

E19 Why does benzene take electrophilic substitution readily but nucleophilic substitution only with difficulty?

E20 How would you turn each of these into benzene? (i) Ethyne; (ii) Ethene; (iii) Hexane

E21 Draw every alkene that hydrogenates to 2-methylbutane.

E22 Rank each set by falling reactivity towards an electrophile E^+: (a) Chlorobenzene, 2,4-dinitrochlorobenzene, *p*-nitrochlorobenzene; (b) Toluene, *p*-$H_3C–C_6H_4–NO_2$, *p*-$O_2N–C_6H_4–NO_2$.

E23 Of benzene, *m*-dinitrobenzene and toluene, which nitrates most readily, and why?

E24 Anhydrous aluminium chloride aside, which Lewis acid would serve for ethylating benzene?

E25 Why is the Wurtz reaction a poor choice for making an alkane with an odd number of carbons? Show why with one example.